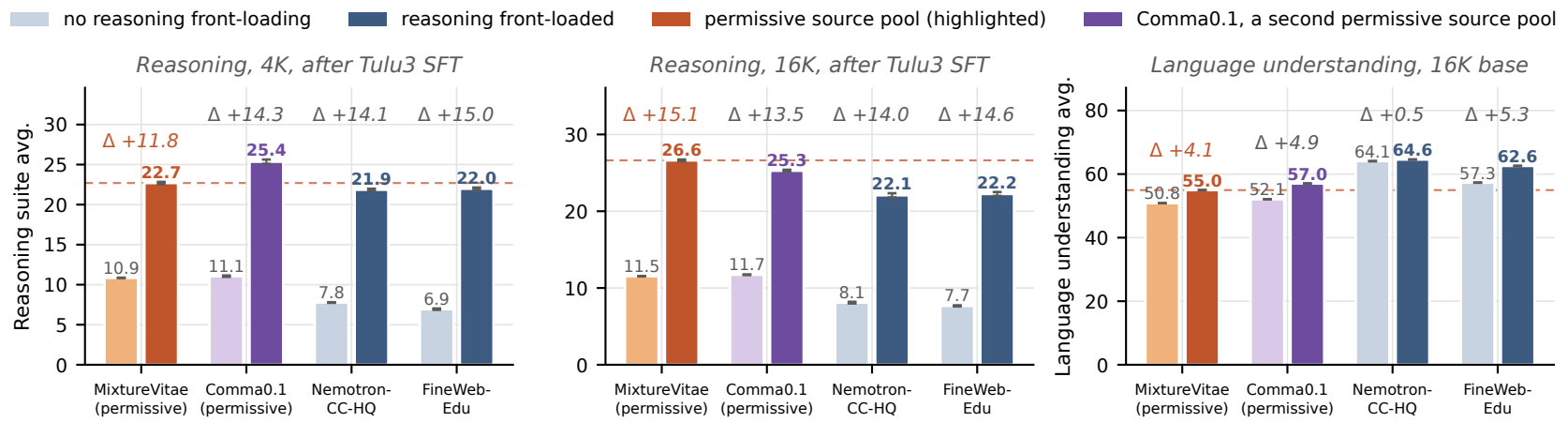
no reasoning front-loading
reasoning front-loaded
permissive source pool (highlighted)
Comma0.1, a second permissive source pool
Reasoning, 4K, after Tulu3 SFT
Reasoning suite avg.
Δ +11.8
Δ +14.3
Δ +14.1
Δ +15.0
22.7
25.4
21.9
22.0
10.9
11.1
7.8
6.9
MixtureVitae (permissive)
Comma0.1 (permissive)
Nemotron-CC-HQ
FineWeb-Edu
Reasoning, 16K, after Tulu3 SFT
Δ +15.1
Δ +13.5
Δ +14.0
Δ +14.6
26.6
25.3
22.1
22.2
11.5
11.7
8.1
7.7
Language understanding, 16K base
Language understanding avg.
Δ +4.1
Δ +4.9
Δ +0.5
Δ +5.3
55.0
57.0
64.6
62.6
50.8
52.1
64.1
57.3